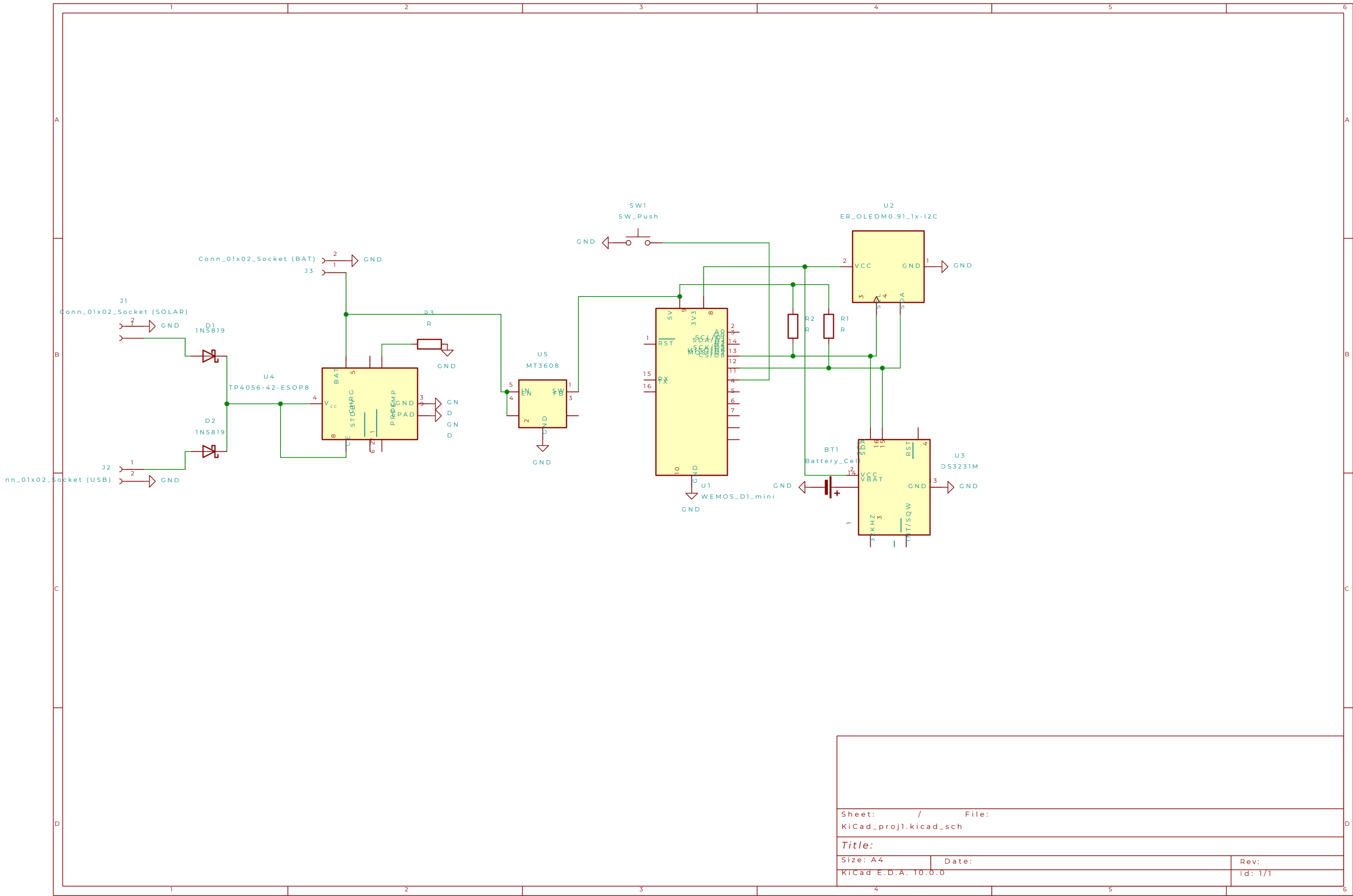


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Physical Office Event Timer Clock

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[GitHub](#)



How it works

logic	The Wemos D1 Mini acts as the "brain," calculating the time difference between the current moment and the saved event timestamp
Timekeeping	A high-precision DS3231 RTC module provides the time via the I2C bus, ensuring accurate tracking even during power failures
Storage	The event start time is stored in the EEPROM (permanent memory), so the timer doesn't reset if the battery dies
Display	The OLED screen continuously updates to show the elapsed time in a user-friendly format (Years, Months, Days, etc.) .
Power	The device is powered by a dual-input system: 5V USB for primary use and a Solar Panel for trickle-charging a Li-ion battery buffer

Challenges and Solutions

Drift and Inaccuracy

Integrated an external DS3231 RTC module, which is temperature-compensated and keeps accuracy within seconds per year.

Power Outages

EEPROM Storage & RTC Backup: Used permanent memory for the start date and a coin-cell battery (CR2032) to keep the clock alive during main battery failure.

Manual Configuration

Wi-Fi Web Interface: Created a local Access Point ("My_Event_Timer") that allows any smartphone to set the date via a browser-based date picker.

Ideas for future improvements

NTP Auto-Sync	Update firmware to connect to office Wi-Fi and automatically sync time with Global Time Servers for "zero-maintenance" accuracy
Centralized Monitoring	Implement an MQTT protocol to send "uptime" or "event" data to a company dashboard (e.g., a "Safety Leaderboard" for all office floors)
Enclosure Aesthetics	Transition from a 3D-printed plastic case to a more premium laser-cut wood or brushed aluminum finish to match modern office interiors
Remote Web Management	Move from a local Access Point to a cloud-based management system for remote reset and status monitoring

Thanks